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10/19/24
Rheology 596
Critique 1

Analysis of Aihui Xiu et. al.'s "Rheological properties of Salecan as a new source of thickening agent"

This research paper aims to examine the rheology of Salecan, a novel soluble glucan produced by Agrobacterium sp. ZX09. The purpose of this research is to determine the glucan's feasibility as a thickening agent in the food industry. The team tested the glucan at several different concentrations, being 0.3%, 0.5%, 1.0% and 1.5% (w/w). These solutions were tested to determine steady flow properties at different shear rates, viscosity at different temperatures, and the effect of heating, cooling, and pH has on viscosity. While thorough in their analysis, I believe that the team experienced several issues with their experiment, which led to erroneous data.

Firstly, their experiment producing the viscosity across differing shear rates shows that Salecan shows a linear viscoelastic regime for a shear rate of 0 to 0.007 s^{-1} for the tested solution concentrations. After that, the viscosity decreases linearly with respect to shear rate. This is sensible and expected for a solution of polysaccharides, and proved a basis for the rest of the experiment, as it proves that Salecan experiences shear thinning at higher shear rates.

The experiment goes on to test different solutions of Salecan at increasing temperatures, which is where I presume their first issue occurred. The graphs show that for all concentrations, viscosity increases as temperature increases, supporting the conclusion that the fluid experiences a gel-like nature for increasing temperatures. In the figure showing their data, it shows a steep decline in viscosity after 60 ∘C. This is counter intuitive and would suggest that for their testing apparatus (a plate geometry sensor with a 0.05 mm gap), severe wall slip occurred as the center of the solution increased in viscosity. The team attributed this change to the "critical phenomena theory", a theory that explains the dramatic shift in viscosity to a critical temperature during which there is a phase change. While this is a plausible explanation, the experiment should be performed with other geometries that account for wall slipping to further conclude whether this is what is happening, as opposed to severe wall slipping.

The team continued to test other rheological properties of Salecan, testing the effect that heating and freezing had on the solution. This experiment resulted in more consistent results, showing that for solutions of 0.3% and 0.5% Salecan, heating the solution prior to testing resulted in a lower viscosity for low shear rates, eventually matching normal solutions at shear rated of 50 s^{-1} or higher. The team also tested the effect that pH had on the solution, showing that the viscosity of a solution of 1.0% Salecan remained relatively stable between 6 and 12, increasing below 4 and above 12.

The other error in this experiment that I concluded occurred during testing the dynamic viscoelastic properties of the solution. I want to draw attention to two graphs in particular, the graph showing G' and G'' vs frequency and the graph showing the loss factor tanδ vs frequency (ω). In these graphs, the solutions of 0.5%, 1.0%, and 1.5% follow a similar trend. For G' and G'', both factors increase slowly as ω increases. Yet for the solution of 0.3% Salecan, the storage modulus (G’) increases dramatically at higher frequencies, overpassing the solution of 0.5% and preparing to overtake 1.0% and 1.5%. This alone would be strange, but is double strange in conjunction with the second graph, showing the loss factor vs frequency. In this graph, the solution of 0.3% starts at a very high loss factor, much higher than the other samples and decreases slowly for increasing frequencies. At around 20 rad/s, the loss factor decreases sharply, plunging below the other solutions. Both of these graphs suggest that a solution of 0.3% Salecan experiences a sharp transition into an elastic solid at higher frequencies, while more concentrated solutions do not. This does not match the conclusions of earlier experiments, which show that a 0.3% solution of Salecan acts similar to other similar concentrations. The higher storage modulus at 50 rad/s implies that a solution of 0.3% Salecan occurs immense shear thickening at higher frequencies, while we know all solutions should exhibit shear thinning. There are several factors that could have contributed to these errors, primarily the preparation of the solution and the geometry used to test the increasing frequency for dynamic properties.

Overall, the researchers made the correct conclusions, stating that a solution of Salecan exhibits shear thinning and pseudoplasticity, while showing little change in viscosity as a function of temperature under 55 degrees C. This of course neglects the erroneous data they gathered in showing that at high temperatures the solutions experienced extremely low viscosities and that at high frequencies and low concentrations a solution of Salecan exhibits extremely viscoelastic properties. In conclusion, several erroneous experiments could lead to making incorrect conclusions, though a more thorough analysis would lead to more scrutiny in the findings.